Annotation Convention, Not Architecture, Limits Cross-Facility Detection of Electric Vehicle Battery Components

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Abstract:
Automated disassembly of end-of-life electric vehicle battery packs depends on a detector that locates modules and busbars across the pack types a recycling line actually receives. Published detectors are trained and validated at a single facility. This paper assembles a cross-facility corpus from 13 independent public sources spanning ten pack families and disaggregates accuracy by source. Error proves concentrated rather than spread: per-source module accuracy spans 1.000 to 0.097, and the collapse occurs where a source disagrees about what the target class denotes rather than where the imagery is difficult. A screening procedure is proposed that identifies such sources from a single training run using a robust outlier criterion, together with a training-free variant computed from annotation geometry alone. Both pass a positive control in which a consensus source relabelled to a sub-component convention is recovered while no other source changes status. Two further results concern method. Under a matched training budget, self-supervised pretraining is marginally worse on conventionally annotated packs and better by a factor of 3.5 under annotation shift. And validation data drawn from a single divergent source inverts checkpoint selection, costing 51% relative accuracy in both architectures tested. The corpus and evaluation code are released.

1 INTRODUCTION

Electric vehicles are reaching end of life (EoL) in large numbers, and the battery packs they carry are simultaneously a hazard and a source of critical raw materials. Lithium, cobalt and nickel are all classified by the European Commission as critical raw materials with elevated supply risk (European Commission, 2023; Harper et al., 2019), and European recycling capacity demand is projected to reach 170 GWh by 2035 (Transport and Environment, 2024). Regulation (EU) 2023/1542 additionally requires documented condition data across the battery lifecycle (European Parliament and Council, 2023), which manual inspection delivers inconsistently.

Disassembly of EoL packs nevertheless remains predominantly manual. The operation is labour intensive, slow, and exposes workers to high voltage and to the risk of thermal runaway. A substantial research effort is consequently directed at automating it (Zang et al., 2024; Tan et al., 2025).

Every such system begins with perception. Before a manipulator can unfasten, grasp or lift a component, the battery modules and the busbars that connect them must be located in the image. The established approach is a supervised object detector trained on imagery collected at a single facility or from a single experimental rig. This practice conceals a problem specific to the application. A recycling line does not receive a single pack type; it receives whatever arrives on a given day, from different manufacturers, in different states of disassembly, under whatever illumination the building provides. Accuracy measured on the distribution a detector was trained on therefore carries limited information about the next pack through the door.

To quantify this effect a cross-facility benchmark is required, and none exists for the task. Every study surveyed in Section 2 collected and annotated its own imagery, and none released it. To address this gap, this paper assembles a corpus of 4,425 training images from 13 independent public sources spanning ten EV pack families, constructs a held-out cross-facility benchmark of 225 images containing 780 instances, and evaluates detectors on it under a single harmonised protocol.

The principal finding is that the dominant failure mode in this setting is neither image difficulty nor model capacity. It is disagreement between sources about what the target class denotes. Per-source module accuracy spans 1.000 to 0.097, and inspection confirms that the source on which accuracy collapses annotates small internal sub-components as modules while every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently, so the collapse constitutes a disagreement about the target rather than a failure of perception.

This observation is developed into a screening procedure. The contributions of this paper are as follows:

1. A cross-facility corpus and benchmark for EV battery module and busbar detection, assembled from 13 independent public sources spanning ten pack families, verified free of train-test leakage by perceptual hashing and released with a per-source provenance and licence record.
2. A convention-consistency screening procedure that identifies annotation-divergent sources from a single merged-corpus training run using a robust outlier criterion on per-source disaggregated accuracy, together with a positive control in which a consensus source relabelled to a sub-component convention is recovered by the screen while no other source changes status.
3. A training-free variant of the same procedure, computed from annotation geometry alone over 4,760 label files, which identifies the same source with no training run and permits a corpus to be screened before a detector is trained.
4. An architecture comparison under a matched 60-epoch budget, establishing that self-supervised pretraining is marginally worse on conventionally annotated packs, at 0.955 against 0.979, and substantially better under annotation shift, at 0.337 against 0.097, a factor of 3.5.
5. The finding that validation data drawn from a single divergent source inverts checkpoint selection rather than merely biasing it: the selected checkpoint is worse than the fixed-budget final checkpoint by 0.235 mAP@50, a relative loss of 51%, with non-overlapping bootstrap intervals and an effect 80 times the seed-to-seed spread.
6. Five negative results, covering open-vocabulary detection, automatic relabelling, naive multi-source merging, synthetic augmentation and prolonged fine-tuning of a pretrained model, each of which constrains the design space.

Relation to prior work by the same authors. The single-source specialist evaluated in this paper is the detector reported in (Katiyar et al., 2026), which addressed detection and condition assessment at a single facility using a CPU-deployable two-stage pipeline. That work could not test cross-facility transfer, which is precisely the limitation the present paper addresses. The corpus, the benchmark, both screening procedures and all evaluation reported here are new, and the condition-assessment stage of the earlier work is deliberately not revisited.

2 RELATED WORK

2.1 Perception for Battery Disassembly

Zang et al. surveyed robotic EV battery disassembly and concluded that the process remains predominantly manual, with perception and task uncertainty identified as the principal obstacles (Zang et al., 2024). Wegener et al. demonstrated that pack designs vary sufficiently between manufacturers that fixed automation does not transfer, which is the practical reason perception must generalise (Wegener et al., 2014). However, that study addressed mechanical disassembly sequencing and did not evaluate a perception component.

Most published perception work for this application targets fasteners rather than components. Li et al. proposed a screw detection method combining Faster R-CNN with rotation edge similarity, reporting improved localisation of the screw axis (Li et al., 2021). However, the method was validated on a single collection of laboratory imagery. Li et al. developed a YOLOX-family detector for active screw detection in EV battery disassembly using 1,719 hand-annotated images (Li et al., 2023). However, the dataset was not released and accuracy was reported only in domain. Naseri et al. presented a comparison of YOLOv11 against Faster R-CNN for the same task, and observed that the field is hampered by a lack of public datasets; the study resorts to laptop screws as a proxy domain, which limits the conclusions that transfer to battery imagery (Naseri et al., 2026).

Module-level and busbar-level perception is comparatively under-explored. Tan et al. developed a collaborative robot that unfastens and collects screws by fusing vision with force and torque sensing, reporting above 90% unfastening efficiency (Tan et al., 2025). However, the vision component operates on a single fixed pack geometry, so the reported figure does not characterise transfer. Gerlitz et al. applied instance segmentation with point-cloud registration in order to grasp busbars on an industrial module (Gerlitz et al., 2023). However, the approach is restricted to one module type and requires a reference model of that type.

A recurring theme is data scarcity. Every study named above collected and hand-annotated its own imagery, and none released it. That absence motivates the benchmark presented in this paper.

2.2 Cross-Dataset Evaluation

Domain shift is a well-documented failure mode in visual recognition. Recht et al. demonstrated that classifiers which appear strong on the original ImageNet test split lose accuracy on a replication set collected by the same protocol (Recht et al., 2019). However, that study varied the sampling procedure while holding the label taxonomy fixed. Koh et al. assembled the WILDS benchmark of in-the-wild distribution shifts across ten application domains, and established that standard training degrades substantially under such shift (Koh et al., 2021). However, WILDS likewise holds the annotation protocol constant within each dataset, whereas the corpus considered in this paper varies both the imaging distribution and the annotation protocol between sources.

Northcutt et al. demonstrated that pervasive label errors in established test sets destabilise benchmark rankings, and estimated an average label error rate of 3.4% across ten widely used datasets (Northcutt et al., 2021). However, that work treats label error as noise about an agreed target, whereas the failure characterised here is systematic disagreement about what the target is. The distinction is consequential for merged corpora: noise averages out with scale whereas a divergent convention does not, and the proposed screening procedure operationalises exactly that distinction.

Penquitt et al. developed Rechecked, a modular framework combining automatic label-error detection with crowd-sourced verification, and applied it to the pedestrian class of KITTI, in which 18% of the original ground-truth labels were found to be missing or inaccurate (Penquitt et al., 2026). However, the framework corrects errors relative to an annotation standard that is agreed within a single dataset, and the authors report that the best available detection methods still miss up to 66% of label errors. The failure characterised in the present work is of a different kind: the sources do not disagree about individual boxes, they disagree about what the class denotes, and consequently no correction procedure operating within a single convention can reconcile them. Section 5.8 reports an attempt at exactly such a reconciliation, and its failure.

Abou Akar et al. investigated multi-source training for industrial object detection, combining real imagery with rendered and generatively synthesised data, and demonstrated that the composition of the training mixture materially affects generalisation (Abou Akar et al., 2026). However, the sources compared in that study share a single annotation standard by construction, because the synthetic data is generated with ground truth attached. The corpus considered here is multi-source in a stricter sense, in that each contributing source carries an independently authored annotation standard.

2.3 Self-Supervised Representations for Detection

Self-supervised backbones such as DINOv2 (Oquab et al., 2024) learn features that transfer without task-specific fine-tuning, and detection transformers (Carion et al., 2020) initialised from them (Roboflow, 2025) are reported to generalise better than detectors trained from scratch in low-data regimes. However, that claim has been established principally on natural imagery with a consistent annotation protocol. Whether it holds on the cluttered, specular imagery of battery internals, and whether it holds under annotation shift specifically, is evaluated in Section 5.5.

2.4 Open-Vocabulary Detection and Automatic Annotation

Open-vocabulary detectors such as YOLO-World (Cheng et al., 2024) and Grounding DINO (Liu et al., 2024) localise objects specified by free-text prompts and have been proposed as a route to annotation-free deployment and to automatic pre-annotation. However, their accuracy depends on the target concept being represented in the vision-language pre-training corpus. The behaviour of both on battery internals is reported in Section 5.8.

Benchmark design for industrial vision has precedent outside this application. Bergmann et al. introduced MVTec AD, which established that a carefully constructed benchmark with documented acquisition conditions can redirect an entire sub-field (Bergmann et al., 2019). However, MVTec AD was collected by one group under controlled conditions and therefore does not exhibit the inter-source annotation disagreement examined here. Zhou et al. surveyed domain generalisation methods and organised them into data manipulation, representation learning and learning strategy (Zhou et al., 2023). However, that taxonomy presumes a fixed label space across domains, which is the assumption the present work shows to fail on assembled corpora.

To the best of our knowledge, no public cross-facility benchmark for EV battery component detection exists, and no prior study has quantified the contribution of annotation-convention divergence to cross-source detection accuracy. Therefore, this research presents both.

3 CORPUS AND BENCHMARK

3.1 Sources

A corpus of 4,425 training images was assembled from 13 independent public sources spanning ten EV pack families, annotated with two classes, module and busbar. Table 1 presents the principal sources with their image counts and pack families.

The images and the polygon annotations originate with those sources. The contribution made in this paper is the assembly, the deduplication, the audit and the benchmark construction, not the annotation itself. Licences differ across sources and one is non-commercial, so per-source provenance is recorded and released so that any single source can be excluded by a downstream user.

Table 1: Principal corpus sources. Thirteen sources in total, spanning ten EV pack families.

Source	Images	Pack family
ev-battery-comp-det.	1,045	multi-manufacturer
ev-battery-components	680	Audi / BMW / Chevrolet
last_exp4	483	multi-manufacturer
ev-battery-pack	435	single laboratory rig
battery_comp	280	Nissan Leaf
ue_d1_defect	237	module close-ups
final_mobilenet	216	multi-manufacturer
automated-disassembly	117	disassembly cell
bmw_i3	111	BMW i3
ue_rav4	96	Toyota RAV4
Others (three sources)	725	mixed

3.2 Deduplication and Leakage Control

Reported accuracy is informative only against a benchmark that does not overlap the training data. All 4,440 candidate images were therefore compared by difference hashing (dHash) at a Hamming distance threshold of 6. That threshold was selected because it separates augmentation-derived near-duplicates from genuinely distinct frames on this corpus; values below 4 admitted rotated duplicates and values above 8 rejected distinct frames of the same pack.

Fifteen same-source near-duplicates were removed. A subsequent cross-source pass at a looser threshold returned zero duplicates, which confirms that the sources are genuinely independent collections rather than republications of one another.

This control proved necessary rather than precautionary. One candidate source, a 712-image public collection covering 19 vehicle types, republishes imagery from another source already present in the corpus: 78 of its images were byte-identical duplicates of training images, a fact stated in its own documentation but easily overlooked. Merging without verification would have placed training images directly into the benchmark and inflated every subsequent measurement.

3.3 Annotation Audit

All 4,425 training images were audited for label and image quality. Table 2 presents the outcome. Of the corpus, 23.5% carries no labels, 12.3% scores low on a Laplacian-variance blur metric,

Table 2: Annotation and image-quality audit of the 4,425-image training corpus.

Flag	Share	Assessment
Unlabelled	23.5%	partly harmful
Low blur score	12.3%	benign
Greyscale	20.4%	benign
Degenerate boxes	0.0%	—
Polygon annotations	86.8%	16,945 of 19,522

20.4% is greyscale, and no image contains a degenerate box.

Two findings changed how the corpus was treated. Firstly, the blur and greyscale flags are largely false alarms. The lowest-scoring images are smooth, closed pack lids, for which a Laplacian-variance metric reads flat metal as blurred, and the greyscale images originate entirely from two monochrome sources. Both categories add useful variation to the corpus, so deletion on the metric alone would degrade it.

Secondly, the harmful category is unlabelled images that nonetheless contain visible components. A proportion of the 23.5% is legitimate background imagery, which is beneficial in moderation because it suppresses false positives. However, inspection confirmed open packs with clearly visible busbar assemblies carrying no annotations at all. Such frames actively teach the detector that busbars are background, and they are a plausible contributor to the weak busbar accuracy reported in Section 5.5.

The audit also established that 16,945 of the 19,522 annotations are polygon masks, with between 5 and 151 vertices, rather than axis-aligned boxes. These are converted to enclosing boxes for detection training and evaluation. It should be noted that an early version of the evaluation pipeline silently discarded every polygon-format label, trained on 20% of the available annotations, and produced an inflated score that appeared plausible on inspection. The failure is easy to introduce and difficult to detect, and ground-truth instance counts should be reconciled across pipelines as a matter of routine.

3.4 Evaluation Sets

Two evaluation sets are used and are kept distinct throughout. A cross-facility benchmark of 225 images containing 780 instances was drawn from all sources and held out entirely from training; it supports the single-source comparison of Section 5.1 and nothing else. The screening procedures of Sections 4.3 and 4.4, and every result that depends on them, operate instead on a stratified sample of 1,628 images spanning 11 sources drawn from the corpus itself, for the reason given in Section 4.3.

4 METHOD

4.1 Detectors and Training Configurations

Three configurations are compared.

The single-source specialist is a YOLOv8n detector trained only on the single-laboratory source at an input resolution of 768 px, following the conventional approach in this literature and reproducing the setting under which published accuracy figures for this task are typically obtained.

The multi-source YOLO11n configuration (Jocher and Qiu, 2024) comprises 2.6 M parameters and was trained from COCO initialisation (Lin et al., 2014) on all 13 sources at 640 px with stochastic gradient descent.

The multi-source RF-DETR configuration (Roboflow, 2025) is a real-time detection transformer initialised from the published RF-DETR-Nano checkpoint, whose backbone derives from self-supervised DINOv2 pretraining (Oquab et al., 2024), fine-tuned end to end on the same data at 672 px. All 30.6 M parameters are updated; the backbone is not frozen.

Matched training budget. An architecture comparison is meaningful only when both models receive equivalent training. Both multi-source configurations are therefore trained for exactly 60 epochs, at the closest input resolutions the two architectures accept, since RF-DETR requires a multiple of 56 and 672 is the nearest such value to 640. Early stopping is disabled and the final checkpoint is taken, for the reason established in Section 5.6: the validation data available for this corpus is drawn from a single source, and selecting an epoch against it inverts the ordering of checkpoints. YOLO11n was trained with three random seeds; RF-DETR was trained once at the full budget, a limitation stated in Section 7.

4.2 Harmonised Evaluation Protocol

Confidence scores are not comparable across architectures, and mixing metric implementations produced a spurious result in early experiments conducted for this study. Every model is therefore scored with a single metric implementation, at a common confidence threshold of 0.05, on identical images, against identical ground truth including polygon-derived boxes. The threshold of 0.05 was selected because it lies below the operating point of all three detectors and therefore does not truncate the precision-recall curve for any of them.

4.3 Convention-Consistency Screening

The proposed screening procedure identifies sources whose annotation convention diverges from the corpus consensus, using a single merged-corpus training run and no additional annotation.

Let $\mathcal{S} = \{S_1, \ldots, S_n\}$ denote the sources contributing to the benchmark, and let a_i denote the class-specific mAP@50 obtained by the merged-corpus detector on the held-out images of source S_i. A source is flagged as convention-divergent when its accuracy falls below a robust lower fence, which is defined as follows:

$$a_i < \mathrm{med}(\mathbf{a}) - k \cdot c \cdot \mathrm{mad}(\mathbf{a}), \tag{1}$$

where $\mathbf{a} = (a_1, \ldots, a_n)$ denotes the vector of per-source accuracies, med denotes the median, mad denotes the median absolute deviation, $c = 1.4826$ rescales the median absolute deviation to a consistent estimator of the standard deviation under normality (Rousseeuw and Croux, 1993), and k denotes a sensitivity parameter. A value of $k = 3$ is used throughout, which is the conventional choice for outlier rejection and which is shown in Section 5.2 to separate the divergent source from the consensus by a wide margin on this corpus.

The median and the median absolute deviation are used in place of the mean and the standard deviation because a divergent source inflates both of the latter and thereby partially masks itself. Consequently, the proposed procedure remains sensitive when the divergent source is large, which is the case on this corpus.

Two practical conditions apply. Firstly, a source must contribute enough instances for its accuracy to be estimated stably; a floor of 100 module instances is used throughout, below which a source is excluded from the screen rather than reported. An earlier evaluation at 40 instances produced a swing of 0.36 mAP@50 on the smallest source between two models that differed only in the labels of an unrelated source, which is sampling noise rather than signal. Secondly, the screen is computed on a stratified sample of the corpus itself rather than on held-out data, because it is a curation diagnostic and not a generalisation measure: a source whose labels contradict the majority convention is not fitted well even during training. Absolute values are consequently optimistic and only the relative ordering is used.

A source flagged by the proposed procedure is not discarded from the corpus. It is reported separately as an out-of-convention benchmark, on the grounds established in Section 5.2 that its labels encode a different task rather than a degraded version of the same one. The procedure requires no additional annotation, no additional training run beyond the merged-corpus model that is trained in any case, and no prior assumption about which source is divergent.

4.4 Training-Free Screening from Annotation Geometry

The procedure of Section 4.3 requires a merged-corpus training run. A cheaper variant operates on the annotation files alone, which permits a corpus to be screened before any detector is trained.

A source that annotates sub-components rather than pack-level modules produces many small boxes per image, whereas a source following the consensus convention produces few large ones. That signature is captured by a granularity index, which is defined as follows:

$$g_i = \frac{n_i}{s_i}, \tag{2}$$

where n_i denotes the mean number of module instances per image in source S_i and s_i denotes the median normalised module scale, taken as the square root of the enclosing-box area in normalised image coordinates. Scale is expressed as a square root so that the index is dimensionally a count per unit length and is therefore invariant to image resolution. The robust criterion of Equation 1 is then applied to $\mathbf{g}$ with the inequality reversed, since convention divergence raises the index rather than lowering it.

5 RESULTS

5.1 The Single-Source Evaluation Gap

Table 3 quantifies the motivating problem. The specialist attains 0.818 mAP@50 on the single-source benchmark on which it was trained and validated, and 0.277 on a cross-facility benchmark

Table 3: Single-source against cross-facility accuracy for the specialist detector, mAP@50.

Model	In-domain	Cross-facility
Single-source specialist	0.818	0.277

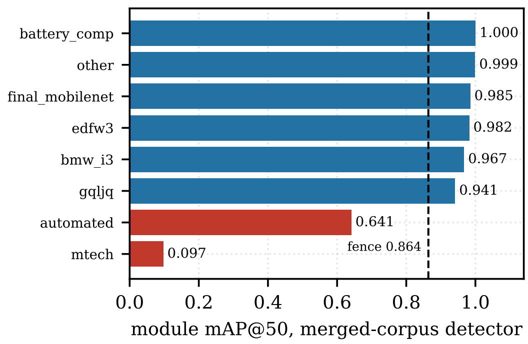

Figure 1: Per-source module accuracy for the merged-corpus detector, with the robust lower fence of Equation 1 at $k = 3$. Two sources fall below the fence, one of them by a wide margin.

drawn from all sources, a relative loss of 66%. Its in-domain accuracy therefore does not predict behaviour elsewhere.

It should be noted at the outset that the in-domain benchmark here is itself the source whose annotation convention is shown below to diverge from the corpus consensus. The figure therefore conflates generalisation loss with the specialist having been trained on an idiosyncratic convention, and it is read as an upper bound on the generalisation component rather than as a measurement of it. The remaining results do not depend on it.

5.2 Per-Source Screening

The merged-corpus detector was evaluated per source on a stratified sample of 1,628 images spanning 11 sources, using the procedure and instance floor of Section 4.3. Figure 1 presents the outcome. Accuracy on six sources lies between 0.941 and 1.000, falls to 0.641 on one source and collapses to 0.097 on another. Applying Equation 1 with $k = 3$ yields a lower fence of 0.864, and two sources fall below it.

The two flagged sources differ in character. The source at 0.097 lies 0.767 below the fence, an order of magnitude below the consensus band. Visual inspection confirms the diagnosis: it annotates small internal sub-components in extreme macro views as modules, whereas every other source annotates pack-level modules. The detector learned the majority convention and applied it consistently, so the collapse constitutes a disagreement about the target rather than a failure of perception. The source at 0.641 lies 0.223 below the fence and is a milder case; its granularity index in Section 5.3 is the second highest in the corpus but does not exceed the training-free fence, so the two screens disagree about it. That disagreement is informative, and indicates the model-based screen to be the more sensitive of the two.

Three sources were excluded for contributing fewer than 100 module instances. One of these annotates busbars exclusively and contributes no module instances at all, a fact the screen surfaces without any manual inspection.

5.3 Training-Free Screening

Equation 2 was evaluated over all 4,760 label files in the corpus, using no images and no trained model. The divergent source attains a granularity index of 40.7, against 15.0 for the next highest source and a median of 9.00 across the corpus. Applying the robust criterion with $k = 3$ yields an upper fence of 29.02, and exactly one source exceeds it.

The source identified is the same one the model-based procedure flags most strongly. The two screens are methodologically independent: one measures how a trained detector behaves on images of each source, and the other measures only the geometry of the annotations, with no model involved. Their agreement provides a second and independent line of evidence that the source encodes a different annotation convention rather than presenting harder imagery.

The practical implication is that a corpus may be screened before a detector is trained. It should be noted that the training-free index detects divergence in annotation granularity specifically, which is the form present in this corpus. Conventions that disagree about class boundaries while preserving object scale, for example a source that labels a cooling plate as a module, would not be separated by Equation 2 and would still require the model-based procedure.

5.4 Positive Control

Both screens were applied to a corpus in which the divergent source was identified after the fact, so neither has been tested against a case whose answer is known in advance. To supply one, a controlled positive was constructed. A consen-

Table 4: Positive control. A consensus source is relabelled to a sub-component convention and the detector is retrained. Values are module mAP@50 for the model-based screen and the granularity index for the training-free screen.

Source	Original	Manipulated	Change
model-based screen, fence 0.864 / 0.815			
battery_comp	1.000	1.000	0.000
edfw3	0.982	0.973	−0.009
bmw_i3	0.967	0.950	−0.017
manipulated	0.941	0.667	**−0.274**
automated	0.641	0.591	−0.050
divergent	0.097	0.097	0.000
training-free screen, fence 29.02 / 35.24			
manipulated	11.1	185.8	**+174.7**

Table 5: Architecture comparison under a matched 60-epoch budget, module mAP@50. Consensus is the mean over the six sources above the fence.

Model	Consensus	Mild outlier	Divergent
YOLO11n	0.979	0.641	0.097
RF-DETR	0.955	0.779	0.337

sus source was relabelled to a deliberately different convention, in which every pack-level module box is replaced by a 3×2 grid of inset subcomponent boxes covering the same region, imitating the granularity of the real divergent source. No image is altered and no busbar annotation is touched, so any screen responding to the manipulation responds to annotation convention and to nothing else. A detector was then retrained on the manipulated corpus under the identical budget.

Table 4 presents the outcome. The manipulated source falls from 0.941 to 0.667 and crosses the fence, while no unmanipulated source moves by more than 0.050 and all but one move by 0.017 or less. The flagged set gains exactly one member and loses none. On the training-free index the same source rises from 11.1, inside the consensus band, to 185.8. Both screens therefore recover a divergence whose ground truth is known by construction, without producing false positives elsewhere.

5.5 Architecture Comparison under a Matched Budget

Both multi-source configurations were trained for 60 epochs and scored per source under the identical protocol. Table 5 presents the comparison.

The result is narrow and specific. On conventionally annotated packs the self-supervised pretrained model is marginally worse, at 0.955 against 0.979. On the mild outlier it scores 0.779 against 0.641, and on the divergent source 0.337 against 0.097, an improvement of a factor of 3.5. Self-supervised pretraining therefore does not detect ordinary modules more accurately in this setting; it degrades considerably more gracefully when the definition of the target moves. On a line receiving heterogeneous packs, graceful degradation is the property of practical value, and the proposed screening procedure and the choice of pretraining are complementary rather than alternative responses to the same problem.

Both architectures produce the same partition under the screen, flagging the same two sources, which indicates the screening procedure to be independent of the detector it is applied to.

Table 6: Effect of selecting a checkpoint against single-source validation data, mAP@50.

Checkpoint	All sources	Module, 95% CI
Val-selected	0.461 ± 0.018	0.552 [0.471, 0.626]
Fixed-budget final	$\mathbf{0.696 \pm 0.003}$	0.735 [0.659, 0.808]

5.6 Single-Source Validation Inverts Checkpoint Selection

A result emerged from the matched-budget runs that was not sought and that bears directly on the paper's central claim. The validation data available for this corpus is drawn overwhelmingly from the divergent source. During training, accuracy on that validation split peaked early and then fell by roughly a factor of five, while both training losses decreased monotonically throughout. Selecting a checkpoint on that signal is therefore selecting for the divergent convention.

The consequence is quantified in Table 6 and Figure 2. Evaluated across all sources, the checkpoint selected by validation accuracy is worse than the final checkpoint by 0.235 mAP@50, a relative loss of 51%. Across three random seeds the final checkpoint attains 0.696 ± 0.003 and the selected checkpoint 0.461 ± 0.018, so the effect is roughly 80 times the seed-to-seed spread. On the module class, 500 bootstrap resamples give 95% intervals of $[0.659, 0.808]$ and $[0.471, 0.626]$, which do not overlap.

The same inversion appears in the self-supervised pretrained model, whose validation signal identified epoch 3 as best and epoch 60 as substantially worse, while across sources the epoch-60 checkpoint scores 0.955 against 0.719 on consensus sources and 0.337 against 0.079 on the divergent source. The effect is therefore a

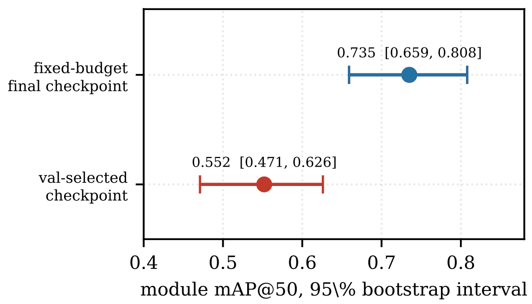

Figure 2: Model selection against single-source validation data. Bars are 95% bootstrap intervals over 500 resamples. The checkpoint chosen by validation accuracy is the worse of the two and the intervals do not overlap.

property of single-source validation and not of one architecture.

Two caveats bound the claim. The evaluation sample is drawn from the corpus rather than held out, so later checkpoints are favoured by memorisation and the absolute size of the gap is optimistic. What memorisation does not explain is the pattern: memorising would raise the final checkpoint uniformly, whereas it gains on the consensus sources while losing on the divergent one, which is a convention trade-off rather than a fitting effect. Secondly, the finding concerns validation data drawn from a single divergent source and does not generalise to well-constructed validation splits.

5.7 Deployment Cost and Zero-Shot Transfer

Evaluated on 17 pack families absent from the corpus, drawn from an unlabelled public collection, the merged-corpus detector attains a mean detection rate of 0.76 and finds busbars in 16 of the 17, with 1.00 on BMW i4 and Hyundai Ioniq against 0.27 on a Volvo truck pack. Module geometry therefore transfers to unseen passenger-vehicle packs, whereas unusual form factors remain the weak axis. It should be noted that a detection-rate proxy measures recall only and cannot detect false positives.

Dynamic INT8 quantisation to the ONNX format reduced median CPU latency from 56.3 ms to 44.8 ms and file size from 6.3 MB to 3.4 MB, for a loss of 0.007 mAP@50, measured as the median over 20 images on an Apple M1 processor with no graphics acceleration. The detector is therefore deployable without a graphics processor at approximately 22 frames per second.

5.8 Negative Results

Five failures constrain the design space for this task. Each is reported because it was expected to succeed and because the expectation is common in the surrounding literature.

Open-vocabulary detection is unusable for this task. YOLO-World (Cheng et al., 2024), prompted with ev battery module and busbar, attained 0.004 mAP@50. Battery internals lie too far outside the vision-language pre-training distribution for the prompts to ground, so the annotation-free route is not currently available for this application.

Naive multi-source merging degrades accuracy. Training on the merged corpus without screening reduced module mAP@50 to 0.094 on the single-source test split, against a single-source baseline of 0.818. More data made the model worse. This result motivated the screening procedure of Section 4.3 and illustrates that curation dominates volume when conventions conflict.

Automatic relabelling cannot reconcile a divergent convention. Reconciliation of the flagged source with the consensus convention was attempted using Grounding DINO (Liu et al., 2024). On clean frames the model recovered two of approximately five modules, and on the greyscale macro imagery that dominates the source it produced boxes covering half the frame. Relabelling 1,756 images on that basis would have injected substantial noise. Convention conflicts are therefore not cheaply repairable after collection, which argues for agreeing annotation guidelines before collection.

Copy-paste and glare augmentation degrade the detector. Copy-paste compositing (Ghiasi et al., 2021) combined with a synthetic glare pass was applied in order to address the specular highlights characteristic of battery internals. Validation mAP@50 declined monotonically over the first six epochs, from 0.503 to 0.270 against a baseline of 0.818, and training was terminated. The likely cause is label noise introduced at paste boundaries together with glare intensity that does not match the statistics of the real bright-condition imagery.

Prolonged fine-tuning degrades the self-supervised pretrained model. Under the matched 60-epoch budget the pretrained model reached its best validation accuracy at epoch 3 or 4 in both runs and improved on no subsequent epoch, ending 27% below its peak. A model that arrives already fitted to the task gains nothing from a budget chosen to suit a model trained from scratch, which is a consideration when matching budgets across architectures.

6 DISCUSSION

Three practical implications follow from the results presented above.

Firstly, when a corpus is assembled from heterogeneous sources, per-source disaggregated reporting is a requirement rather than an option. An aggregate figure computed across conventions that disagree describes no realisable deployment condition, and on this corpus it would average a source at 1.000 with one at 0.097. The proposed screening procedure makes the necessary partition explicit and reproducible, at the cost of a single evaluation pass over a model that is trained in any case, or of no training run at all in its geometry-based form.

Secondly, curation and the choice of pretraining address the same failure mode by different means, and they are complementary rather than alternative. Excluding the divergent source removes it from the aggregate entirely, whereas self-supervised pretraining raises accuracy on it from 0.097 to 0.337 without removing it. A divergent source that has not been identified is therefore still handled more gracefully by the pretrained model, which argues for applying both.

Thirdly, and least expected, the same annotation disagreement that distorts reported accuracy also distorts model selection, and does so invisibly. A practitioner following the default behaviour of a standard training framework would have selected the checkpoint that is worse by 51% relative, while the training logs showed monotonically decreasing losses throughout. Validation splits assembled from heterogeneous corpora therefore warrant the same per-source scrutiny as test splits, and the screening procedures presented here apply unchanged to them.

7 LIMITATIONS

The 66% relative loss reported in Section 5.1 is measured against an in-domain benchmark that is itself the convention-divergent source. The figure therefore conflates generalisation loss with the specialist having been trained on an idiosyncratic convention, and it is read as an upper bound on the generalisation component rather than as a measurement of it. A second in-domain reference source would separate the two contributions. No other result in this paper depends on it.

Both screens are computed on a stratified sample of the corpus rather than on held-out data, because the validation split available here spans only two sources and cannot support a per-source outlier test. Absolute values are consequently optimistic, and later checkpoints are additionally favoured by memorisation. Only relative orderings are used, and the inversion of Section 5.6 rests on a pattern that memorisation does not produce, but a held-out multi-source split would give a cleaner measurement of the magnitudes.

YOLO11n was trained with three random seeds, giving a standard deviation of 0.003. The self-supervised pretrained model was trained once at the full budget, for reasons of compute; a second run reproduced its early-peak behaviour but was halted at epoch 9. Its numbers therefore carry no variance estimate.

The proposed screening procedure requires that divergent sources are a minority of the corpus, since the median defines the consensus. It has been validated on a corpus containing one severely divergent source and one mild case, and would require modification for a corpus in which two or more mutually incompatible conventions are each well represented. It also requires a minimum number of instances per source; three sources in this corpus fall below that floor and are excluded rather than reported.

The benchmark inherits the annotation conventions of its sources and no independently relabelled gold subset was constructed, so model error and label error cannot be separated on it. This is a material limitation for a study whose central claim concerns annotation quality, and its removal is the first item of future work.

The corpus is assembled from public sources rather than collected under controlled conditions, so pack-type coverage is uneven and licences vary. One source representing 24% of the training images is licensed CC BY-NC-SA 4.0, whose Share-Alike condition propagates to any derivative corpus. One further source became unavailable during this study, which is a reproducibility risk inherent to corpora assembled from public repositories.

Finally, absolute accuracy on the busbar class remains modest and is not sufficient for autonomous operation on that class. The results presented here characterise the failure modes of the task rather than deliver a deployable detector.

8 CONCLUSIONS

This paper presented a cross-facility corpus and benchmark for electric vehicle battery component detection, and used it to establish that annotation convention, rather than architecture, is the dominant failure mode in this application. Per-source accuracy for a single detector spans 0.097 to 1.000, and the collapse occurs where a source disagrees about what the target class denotes rather than where the imagery is hard.

The proposed screening procedure identifies such sources from a single training run, and a training-free variant computed from annotation geometry identifies the same source with no training at all. Both pass a positive control in which a consensus source is relabelled to a deliberately different convention: the manipulated source is recovered by both screens while no other source changes status.

Two results about method rather than data follow. Under a matched training budget, self-supervised pretraining is marginally worse on conventionally annotated packs and better by a factor of 3.5 under annotation shift, which narrows the circumstances in which it should be preferred to graceful degradation rather than accuracy. And validation data drawn from a single divergent source inverts checkpoint selection: the selected checkpoint is worse than the fixed-budget final checkpoint by 51% relative, with non-overlapping bootstrap intervals, and the same inversion appears in both architectures tested.

Taken together, these results indicate that dataset design, and specifically agreement on annotation convention before collection, is the first thing to get right in this application, and that the same disagreement propagates silently into model selection if the validation split is not itself checked.

Future work will construct an independently relabelled gold subset in order to separate model error from label error, will repeat both screens on a held-out multi-source split, and will extend the training-free index to forms of convention divergence that preserve object scale.

DATA AVAILABILITY

The benchmark definition, the per-source provenance and licence record, the dataset audit tooling, the evaluation code and the trained detector weights are available at https://github.com/Komil-jon/ev-battery-vision-pipeline.

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